Kamal Maher

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Research interests: Multi-agent systems • Evals • Interpretability • Biosecurity

Education

2020 – 2025 | **Massachusetts Institute of Technology**, PhD in Computational Biology
Thesis: Fundamental representations of regions and interactions in spatial transcriptomics
Advisor: Xiao Wang

2014 – 2018 | **Cornell University**, BS in Neuroscience

Experience

2026 – | **SecureBio**, Senior Research Fellow (via Pivotal)

- Building a multi-agent system that automatically designs evaluations of biological capabilities in frontier models.

2025 – 2026 | **SPAR**, Research Fellow

- Built an agent that iteratively probes and explains learned representations in large language models, improving feature explanation accuracy by 7.7% over existing methods. [1]
- Demonstrated that a widely-used mechanistic interpretability method is incentivized to learn unfaithful explanations of neural network computation. [2]

2021 – 2025 | **Massachusetts Institute of Technology**, Graduate Researcher

- Developed a spectral graph framework for spatial transcriptomics and applied it to a 1.09M cell mouse brain atlas, revealing transcriptional gradients and regions absent from established anatomical references. [3]

2024 | **Genentech**, Aviv Regev Lab Intern

- Extended the same framework to cancer spatial transcriptomics, identifying tumor regions and cell–cell interactions more efficiently than existing GNN-based approaches.

Select Publications (*equal contribution)

[1] *Multi-shot AutoInterp: Agents Can Explain Complex Features By Refining Explanations.*
K. Maher*, S.E. Schrader*, K. Ayonrinde
ICLR 2026 Workshop on Unifying Concept Representation Learning. 2026

[2] *Cross-Layer Transcoders are Incentivized to Learn Unfaithful Circuits.*
G. Lange, R. Goldstein, **K. Maher**, K. Dearstyne
ICML 2026 Mechanistic Interpretability Workshop. 2026 (Spotlight)

[3] *Spatial Atlas of the Mouse Central Nervous System at Molecular Resolution.*
H. Shi*, Y. He*, Y. Zhou*, J. Huang, **K. Maher**, B. Wang, Z. Tang, S. Luo, P. Tan, M. Wu, Z. Lin, J. Ren, Y. Thapa, X. Tang, K.Y. Chan, B.E. Deverman, H. Shen, A. Liu, J. Liu, X. Wang
Nature. 2023